Отчет по практической работе №1

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Группа: БСБО-16-23

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1. Выполненные команды Docker

1.1 Работа с образами

Поиск образов в Docker Hub

```
alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1 (main)
$ docker search nginx
NAME                                      DESCRIPTION                                     STARS     OFFICIAL
nginx                                     Official build of Nginx.                        21204     [OK]
nginx/nginx-ingress                       NGINX and  NGINX Plus Ingress Controllers fo…   115
nginx/nginx-prometheus-exporter           NGINX Prometheus Exporter for NGINX and NGIN…   50
nginx/unit                                This repository is retired, use the Docker o…   66
nginx/nginx-ingress-operator              NGINX Ingress Operator for NGINX and NGINX P…   3
nginx/nginx-quic-qns                      NGINX QUIC interop                              1
nginx/nginxaas-loadbalancer-kubernetes                                                    1
nginx/unit-preview                        Unit preview features                           0
bitnami/nginx                             Bitnami Secure Image for nginx                  203
bitnamicharts/nginx                       Bitnami Helm chart for NGINX Open Source        3
ubuntu/nginx                              Nginx, a high-performance reverse proxy & we…   140
kasmweb/nginx                             An Nginx image based off nginx:alpine and in…   8
rancher/nginx                                                                             3
linuxserver/nginx                         An Nginx container, brought to you by LinuxS…   236
dtagdevsec/nginx                          T-Pot Nginx                                     0
paketobuildpacks/nginx                                                                    0
vmware/nginx                                                                              3
chainguard/nginx                          Build, ship and run secure software with Cha…   5
gluufederation/nginx                       A customized NGINX image containing a consu…   1
cleanstart/nginx                          Secure by Design, Built for Speed, Hardened …   0
antrea/nginx                              Nginx server used for Antrea e2e testing        0
intel/nginx                                                                               0
circleci/nginx                            This image is for internal use                  2
activestate/nginx                         ActiveState's customizable, low-to-no vulner…   0
docksal/nginx                             Nginx service image for Docksal                 1

alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1 (main)
$ |
```

Скачивание образа

```
alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1 (main)
$ docker pull nginx:alpine
alpine: Pulling from library/nginx
5e7756927bef: Download complete
589002ba0eae: Download complete
6b7b6c7061b7: Download complete
3e2c181db1b0: Download complete
bca5d04786e1: Download complete
955a8478f9ac: Download complete
399d0898a94e: Download complete
6d397a54a185: Download complete
Digest: sha256:1d13701a5f9f3fb01aaa88cef2344d65b6b5bf6b7d9fa4cf0dca557a8d7702ba
Status: Downloaded newer image for nginx:alpine
docker.io/library/nginx:alpine

What's next:
    View a summary of image vulnerabilities and recommendations → docker scout quickview nginx:alpine
```

Просмотр локальных образов

```
alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1 (main)
$ docker images
REPOSITORY                TAG       IMAGE ID       CREATED       SIZE
plantuml/plantuml-server  latest    cd3d67a3150a   10 days ago   844MB
jgraph/drawio             latest    324fc8d66731   2 weeks ago   1.33GB
devprom/alm-app           latest    6b3aecab4185   3 weeks ago   2.59GB
nginx                     alpine    1d13701a5f9f   4 weeks ago   92.5MB
docker/welcome-to-docker  latest    eedaff45e3c7   2 years ago   29.5MB
mariadb                   10.7      9a48ac9f196f   2 years ago   539MB
devprom/math              latest    de9fd3d2e732   6 years ago   1.67GB
```

Просмотр истории слоев образа

```
alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1 (main)
$ docker history nginx:alpine
IMAGE          CREATED       CREATED BY                                      SIZE      COMMENT
1d13701a5f9f   4 weeks ago   RUN /bin/sh -c set -x     && apkArch="$(cat …   51.8MB    buildkit.dockerfile.v0
<missing>      4 weeks ago   ENV ACME_VERSION=0.3.1                          0B        buildkit.dockerfile.v0
<missing>      4 weeks ago   ENV NJS_RELEASE=1                               0B        buildkit.dockerfile.v0
<missing>      4 weeks ago   ENV NJS_VERSION=0.9.5                           0B        buildkit.dockerfile.v0
<missing>      4 weeks ago   CMD ["nginx" "-g" "daemon off;"]                0B        buildkit.dockerfile.v0
<missing>      4 weeks ago   STOPSIGNAL SIGQUIT                              0B        buildkit.dockerfile.v0
<missing>      4 weeks ago   EXPOSE map[80/tcp:{}]                           0B        buildkit.dockerfile.v0
<missing>      4 weeks ago   ENTRYPOINT ["/docker-entrypoint.sh"]            0B        buildkit.dockerfile.v0
<missing>      4 weeks ago   COPY 30-tune-worker-processes.sh /docker-ent…   16.4kB    buildkit.dockerfile.v0
<missing>      4 weeks ago   COPY 20-envsubst-on-templates.sh /docker-ent…   12.3kB    buildkit.dockerfile.v0
<missing>      4 weeks ago   COPY 15-local-resolvers.envsh /docker-entryp…   12.3kB    buildkit.dockerfile.v0
<missing>      4 weeks ago   COPY 10-listen-on-ipv6-by-default.sh /docker…   12.3kB    buildkit.dockerfile.v0
<missing>      4 weeks ago   COPY docker-entrypoint.sh / # buildkit          8.19kB    buildkit.dockerfile.v0
<missing>      4 weeks ago   RUN /bin/sh -c set -x     && addgroup -g 101…   5.56MB    buildkit.dockerfile.v0
<missing>      4 weeks ago   ENV DYNPKG_RELEASE=1                            0B        buildkit.dockerfile.v0
<missing>      4 weeks ago   ENV PKG_RELEASE=1                               0B        buildkit.dockerfile.v0
<missing>      4 weeks ago   ENV NGINX_VERSION=1.29.5                        0B        buildkit.dockerfile.v0
<missing>      4 weeks ago   LABEL maintainer=NGINX Docker Maintainers <d…   0B        buildkit.dockerfile.v0
<missing>      5 weeks ago   CMD ["/bin/sh"]                                 0B        buildkit.dockerfile.v0
<missing>      5 weeks ago   ADD alpine-minirootfs-3.23.3-x86_64.tar.gz /…   9.11MB    buildkit.dockerfile.v0
```

Удаление образа

```
alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1 (main)
$ docker rmi nginx:alpine
Untagged: nginx:alpine
Deleted: sha256:1d13701a5f9f3fb01aaa88cef2344d65b6b5bf6b7d9fa4cf0dca557a8d7702ba
```

Практическое задание

Поиск образа postgres:15-alpine

```
alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1 (main)
$ docker search postgres:15-alpine
NAME                    DESCRIPTION                                     STARS     OFFICIAL
postgres                The PostgreSQL object-relational database sy…   14816     [OK]
cimg/postgres                                                           9
circleci/postgres       The PostgreSQL object-relational database sy…   35
kasmweb/postgres        Postgres image maintained by Kasm Technologi…   6
elestio/postgres        Postgres, verified and packaged by Elestio      2
ubuntu/postgres         PostgreSQL is an open source object-relation…   42
chainguard/postgres     Build, ship and run secure software with Cha…   1
artifacthub/postgres                                                    0
cleanstart/postgres     Secure by Design, Built for Speed, Hardened …   0
corpusops/postgres      https://github.com/corpusops/docker-images/     0
geokrety/postgres       Postgres with postgis + quantile and amqp ex…   0
rootpublic/postgres                                                     0
dockette/postgres       My PostgreSQL image with tunning and preinst…   1
vulhub/postgres                                                         1
wayofdev/postgres                                                       0
uselagoon/postgres                                                      0
pgrouting/postgres       Postgres Docker images with PostGIS and dep…   0
openeuler/postgres                                                      0
clarinpl/postgres                                                       0
supabase/postgres       Unmodified Postgres with some useful plugins…   73
mosipdev/postgres                                                       0
brimstone/postgres      postgres image with traefik-cert support        0
trainlineeurope/postgres Extended version of official Postgres https:…  0
blacklabelops/postgres  Postgres Image for Atlassian Applications       4
fredboat/postgres       PostgreSQL 10.0 used in FredBoat's docker-co…   1
```

Скачивание образа postgres:15-alpine

```
alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1 (main)
$ docker pull postgres:15-alpine
15-alpine: Pulling from library/postgres
34c88efd36aa: Download complete
a27cb555d458: Download complete
610258b635a8: Download complete
2a9c4fd442cb: Download complete
ac1ff2379be8: Download complete
9f6d6f38d9d3: Download complete
8c9e2aa22b4c: Download complete
765f80b9a483: Download complete
aa104f8e5799: Download complete
b53228f7876a: Download complete
Digest: sha256:fceb6f86328c36f2438fae3b851b0cc57c4a7e69a58c866d9ce24281f2cf0c9c
Status: Downloaded newer image for postgres:15-alpine
docker.io/library/postgres:15-alpine

What's next:
    View a summary of image vulnerabilities and recommendations → docker scout quickview postgres:15-alpine
```

Поиск образа golang:1.21-alpine

```
alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1 (main)
$ docker search golang:1.21-alpine
NAME                      DESCRIPTION                                     STARS     OFFICIAL
golang                    Go (golang) is a general purpose, higher-lev…   5109      [OK]
circleci/golang           CircleCI images for Go                          21
bitnami/golang            Bitnami Secure Image for golang                 12
okteto/golang             Development environment for golang              6
antrea/golang             Golang image used by Antrea for building        3
rancher/golang                                                            0
alpinelinux/golang        Build container for golang based on Alpine L…   4
corpusops/golang          https://github.com/corpusops/docker-images/     0
rootpublic/golang                                                         0
goharbor/golang                                                           0
jetbrainsinfra/golang     Golang + custom build tools                     2
pivotalcfreleng/golang    this image is deprecated. Please use gcr.io/…   1
jelastic/golang           An image of the Golang application server ma…   0
webhippie/golang          Docker image for Golang                         1
odpks/golang                                                              0
idoall/golang             golang                                          0
amd64/golang              Go (golang) is a general purpose, higher-lev…   3
arm64v8/golang            Go (golang) is a general purpose, higher-lev…   19
arm32v7/golang            Go (golang) is a general purpose, higher-lev…   26
winamd64/golang           Go (golang) is a general purpose, higher-lev…   1
i386/golang               Go (golang) is a general purpose, higher-lev…   2
cfinfrastructure/golang                                                   0
s390x/golang              Go (golang) is a general purpose, higher-lev…   0
cycloid/golang                                                            0
ppc64le/golang            Go (golang) is a general purpose, higher-lev…   1
```

Скачивание образа golang:1.21-alpine

```
alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1 (main)
$ docker pull golang:1.21-alpine
1.21-alpine: Pulling from library/golang
4f4fb700ef54: Already exists
4579008f8500: Download complete
54bf7053e2d9: Download complete
c6a83fedfae6: Download complete
41db7493d1c6: Download complete
Digest: sha256:2414035b086e3c42b99654c8b26e6f5b1b1598080d65fd03c7f499552ff4dc94
Status: Downloaded newer image for golang:1.21-alpine
docker.io/library/golang:1.21-alpine

What's next:
    View a summary of image vulnerabilities and recommendations → docker scout quickview golang:1.21-alpine
```

Просмотр локальных образов

```
alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1 (main)
$ docker images
REPOSITORY                TAG          IMAGE ID       CREATED        SIZE
plantuml/plantuml-server  latest       cd3d67a3150a   10 days ago    844MB
postgres                  15-alpine    fceb6f86328c   11 days ago    392MB
jgraph/drawio             latest       324fc8d66731   2 weeks ago    1.33GB
devprom/alm-app           latest       6b3aecab4185   3 weeks ago    2.59GB
nginx                     alpine       1d13701a5f9f   4 weeks ago    92.5MB
golang                    1.21-alpine  2414035b086e   19 months ago  337MB
docker/welcome-to-docker  latest       eedaff45e3c7   2 years ago    29.5MB
mariadb                   10.7         9a48ac9f196f   2 years ago    539MB
devprom/math              latest       de9fd3d2e732   6 years ago    1.67GB
```

1.2 Работа с контейнерами

Запуск контейнера alpine в интерактивном режиме и выполнение команд внутри контейнера

```
alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1 (main)
$ docker run -it --name test-alpine alpine:latest sh
Unable to find image 'alpine:latest' locally
latest: Pulling from library/alpine
Digest: sha256:25109184c71bdad752c8312a8623239686a9a2071e8825f20acb8f2198c3f659
Status: Downloaded newer image for alpine:latest
/ # ls -la
total 64
drwxr-xr-x    1 root     root         4096 Mar 10 18:42 .
drwxr-xr-x    1 root     root         4096 Mar 10 18:42 ..
-rwxr-xr-x    1 root     root            0 Mar 10 18:42 .dockerenv
drwxr-xr-x    2 root     root         4096 Jan 27 21:19 bin
drwxr-xr-x    5 root     root          360 Mar 10 18:42 dev
drwxr-xr-x    1 root     root         4096 Mar 10 18:42 etc
drwxr-xr-x    2 root     root         4096 Jan 27 21:19 home
drwxr-xr-x    6 root     root         4096 Jan 27 21:19 lib
drwxr-xr-x    5 root     root         4096 Jan 27 21:19 media
drwxr-xr-x    2 root     root         4096 Jan 27 21:19 mnt
drwxr-xr-x    2 root     root         4096 Jan 27 21:19 opt
dr-xr-xr-x  354 root     root            0 Mar 10 18:42 proc
drwx------    1 root     root         4096 Mar 10 18:42 root
drwxr-xr-x    3 root     root         4096 Jan 27 21:19 run
drwxr-xr-x    2 root     root         4096 Jan 27 21:19 sbin
drwxr-xr-x    2 root     root         4096 Jan 27 21:19 srv
dr-xr-xr-x   13 root     root            0 Mar 10 18:42 sys
drwxrwxrwt    2 root     root         4096 Jan 27 21:19 tmp
drwxr-xr-x    7 root     root         4096 Jan 27 21:19 usr
drwxr-xr-x   11 root     root         4096 Jan 27 21:19 var
/ # exit
```

Запуск контейнера в фоновом режиме

```
alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1 (main)
$ docker run -d --name web-server -p 8080:80 nginx:alpine
6e116758adcf5a1c9cc4984896477c619e316009f9616de27d17d3a95fdefd75
```

Просмотр запущенных контейнеров

```
alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1 (main)
$ docker ps
CONTAINER ID   IMAGE                 COMMAND                  CREATED        STATUS         PORTS                                      NAMES
6e116758adcf   nginx:alpine          "/docker-entrypoint.…"   44 seconds ago Up 44 seconds  0.0.0.0:8080->80/tcp                       web-server
c126116175c1   jgraph/drawio         "/docker-entrypoint.…"   4 days ago     Up 33 minutes  8080/tcp, 8443/tcp                         draw
32f3ea7c2f52   plantuml/plantuml-server "/entrypoint.sh"      4 days ago     Up 33 minutes  8080/tcp                                   plantuml
8b74b976684f   devprom/alm-app:latest "/bin/sh -c '( set -…"  4 days ago     Up 33 minutes  0.0.0.0:80->80/tcp, 0.0.0.0:443->443/tcp   alm-app
a598d034bd0d   devprom/math          "/bin/sh -c '( set -…"   4 days ago     Up 33 minutes  80/tcp                                     math
```

Просмотр всех контейнеров (включая остановленные)

```
alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1 (main)
$ docker ps -a
CONTAINER ID   IMAGE                    COMMAND                  CREATED          STATUS                     PORTS
          NAMES
6e116758adcf   nginx:alpine             "/docker-entrypoint.…"   About a minute ago  Up About a minute          0.0.0.0:8080->80/tcp
          web-server
04148796b234   alpine:latest            "sh"                     2 minutes ago    Exited (0) About a minute ago
          test-alpine
c126116175c1   jgraph/drawio            "/docker-entrypoint.…"   4 days ago       Up 34 minutes              8080/tcp, 8443/tcp
          draw
32f3ea7c2f52   plantuml/plantuml-server "/entrypoint.sh"         4 days ago       Up 34 minutes              8080/tcp
          plantuml
05ca91ed6aad   mariadb:10.7             "docker-entrypoint.s…"   4 days ago       Exited (127) 34 minutes ago 3306/tcp
          alm-db
8b74b976684f   devprom/alm-app:latest   "/bin/sh -c '( set -…"   4 days ago       Up 34 minutes              0.0.0.0:80->80/tcp, 0.0.0.0:443-
>443/tcp   alm-app
a598d034bd0d   devprom/math             "/bin/sh -c '( set -…"   4 days ago       Up 34 minutes              80/tcp
          math
8e43b6d93a55   docker/welcome-to-docker:latest "/docker-entrypoint.…" 15 months ago  Exited (0) 15 months ago
          my-first-container
be24ffffa9b8   docker/welcome-to-docker:latest "/docker-entrypoint.…" 15 months ago  Exited (0) 15 months ago
          eloquent_elgamal
```

Просмотр логов контейнера

```
alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1 (main)
$ docker logs web-server
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
```

```
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2026/03/10 18:43:02 [notice] 1#1: using the "epoll" event method
2026/03/10 18:43:02 [notice] 1#1: nginx/1.29.5
2026/03/10 18:43:02 [notice] 1#1: built by gcc 15.2.0 (Alpine 15.2.0)
2026/03/10 18:43:02 [notice] 1#1: OS: Linux 6.6.87.2-microsoft-standard-WSL2
2026/03/10 18:43:02 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
2026/03/10 18:43:02 [notice] 1#1: start worker processes
2026/03/10 18:43:02 [notice] 1#1: start worker process 30
2026/03/10 18:43:02 [notice] 1#1: start worker process 31
2026/03/10 18:43:02 [notice] 1#1: start worker process 32
2026/03/10 18:43:02 [notice] 1#1: start worker process 33
2026/03/10 18:43:02 [notice] 1#1: start worker process 34
2026/03/10 18:43:02 [notice] 1#1: start worker process 35
2026/03/10 18:43:02 [notice] 1#1: start worker process 36
2026/03/10 18:43:02 [notice] 1#1: start worker process 37
2026/03/10 18:43:02 [notice] 1#1: start worker process 38
2026/03/10 18:43:02 [notice] 1#1: start worker process 39
2026/03/10 18:43:02 [notice] 1#1: start worker process 40
2026/03/10 18:43:02 [notice] 1#1: start worker process 41
2026/03/10 18:43:02 [notice] 1#1: start worker process 42
2026/03/10 18:43:02 [notice] 1#1: start worker process 43
2026/03/10 18:43:02 [notice] 1#1: start worker process 44
2026/03/10 18:43:02 [notice] 1#1: start worker process 45
172.17.0.1 - - [10/Mar/2026:18:49:45 +0000] "GET / HTTP/1.1" 200 615 "-" "curl/8.16.0" "-"
172.17.0.1 - - [10/Mar/2026:18:50:08 +0000] "GET / HTTP/1.1" 200 615 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/145.0.
0.0 Safari/537.36" "-"
2026/03/10 18:50:08 [error] 31#31: *2 open() "/usr/share/nginx/html/favicon.ico" failed (2: No such file or directory), client: 172.17.0.1, server: localhost, request:
"GET /favicon.ico HTTP/1.1", host: "localhost:8080", referrer: "http://localhost:8080/"
172.17.0.1 - - [10/Mar/2026:18:50:08 +0000] "GET /favicon.ico HTTP/1.1" 404 555 "http://localhost:8080/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (
KHTML, like Gecko) Chrome/145.0.0.0 Safari/537.36" "-"
```

Подключение к работающему контейнеру

```
alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1 (main)
$ docker exec -it web-server sh
/ # cat /usr/share/nginx/html/index.html
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, the nginx web server is successfully installed and
working. Further configuration is required.</p>

<p>For online documentation and support please refer to
<a href="http://nginx.org/">nginx.org</a>.<br/>
Commercial support is available at
<a href="http://nginx.com/">nginx.com</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
/ # exit

What's next:
    Try Docker Debug for seamless, persistent debugging tools in any container or image → docker debug web-server
    Learn more at https://docs.docker.com/go/debug-cli/
```

Остановка контейнера

```
alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1 (main)
$ docker stop web-server
web-server
```

Запуск остановленного контейнера

```
alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1 (main)
$ docker start web-server
web-server
```

Удаление контейнера

```
alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1 (main)
$ docker rm web-server
web-server
```

Практическое задание

Запуск контейнера с PostgreSQL и выполнение команды psql в контейнере

```
alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1 (main)
$ docker run --name my-postgres -e POSTGRES_PASSWORD=secretpassword -d postgres:15-alpine
a27530479fb4fec965a479154b262ad9590e7cd37de31fc8b369395eb2510061

alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1 (main)
$ docker exec -it my-postgres psql -U postgres -c "SELECT version();"
                                 version
-----------------------------------------------------------------------------------------
 PostgreSQL 15.17 on x86_64-pc-linux-musl, compiled by gcc (Alpine 15.2.0) 15.2.0, 64-bit
(1 row)


what's next:
    Try Docker Debug for seamless, persistent debugging tools in any container or image → docker debug my-postgres
    Learn more at https://docs.docker.com/go/debug-cli/
```

1.3 Работа с томами

Создание именованного тома

```
alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1 (main)
$ docker volume create my-app-data
my-app-data
```

Просмотр томов

```
alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1 (main)
$ docker volume ls
DRIVER    VOLUME NAME
local     4f7575db990c4a5224730f9eb564137df0384a63ad738125e58ffdb0b6574570
local     b23a7672ef20326a20150199b9a49f4899caccf324177266a8edcac5e48a8df7
local     docker_dbdata
local     my-app-data
```

Информация о томе

```
alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1 (main)
$ docker volume inspect my-app-data
[
    {
        "CreatedAt": "2026-03-10T19:09:45Z",
        "Driver": "local",
        "Labels": null,
        "Mountpoint": "/var/lib/docker/volumes/my-app-data/_data",
        "Name": "my-app-data",
        "Options": null,
        "Scope": "local"
    }
]
```

Запуск контейнера с томом

```
alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1 (main)
$ docker run -d --name postgres-db -e POSTGRES_PASSWORD=secret -e POSTGRES_DB=testdb -v my-app-data:/var/lib/postgresql/data -p 5432:5432 postgres:15-alpine
02aead0223fad67c1e60133767ede5d20a68d1828a37527a9653191970162018

alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1 (main)
$
```

Создание тестовой таблицы

```
alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1 (main)
$ docker exec -it postgres-db psql -U postgres -d testdb -c "CREATE TABLE users (id
SERIAL, name TEXT);"
CREATE TABLE

What's next:
    Try Docker Debug for seamless, persistent debugging tools in any container or image → docker debug postgres-db
    Learn more at https://docs.docker.com/go/debug-cli/
```

Занесение данных в таблицу

```
alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1 (main)
$ docker exec -it postgres-db psql -U postgres -d testdb -c "INSERT INTO users (name) VALUES ('EAL 16');"
INSERT 0 1

What's next:
    Try Docker Debug for seamless, persistent debugging tools in any container or image → docker debug postgres-db
    Learn more at https://docs.docker.com/go/debug-cli/
```

Остановка и удаление контейнера (данные сохранятся в томе)

```
alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1 (main)
$ docker stop postgres-db
postgres-db

alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1 (main)
$ docker rm postgres-db
postgres-db
```

Запуск нового контейнера с тем же томом

```
alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1 (main)
$ docker run -d --name new-postgres -e POSTGRES_PASSWORD=secret -e POSTGRES_DB=testdb -v my-app-data:/var/lib/postgresql/data -p 5432:5432 postgres:15-alpine
92b19b88c7736751b7d34f3c5e08e62dd87b1c68535c0a0ad7df7b32eb840773
```

Проверка, что данные сохранились

```
alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1 (main)
$ docker exec -it new-postgres psql -U postgres -d testdb -c "SELECT * FROM users;"
 id |  name
----+--------
  1 | EAL 16
(1 row)


What's next:
    Try Docker Debug for seamless, persistent debugging tools in any container or image → docker debug new-postgres
    Learn more at https://docs.docker.com/go/debug-cli/
```

Практическое задание

Создание тома для статических файлов

```
alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1 (main)
$ docker volume create my-nginx
my-nginx
```

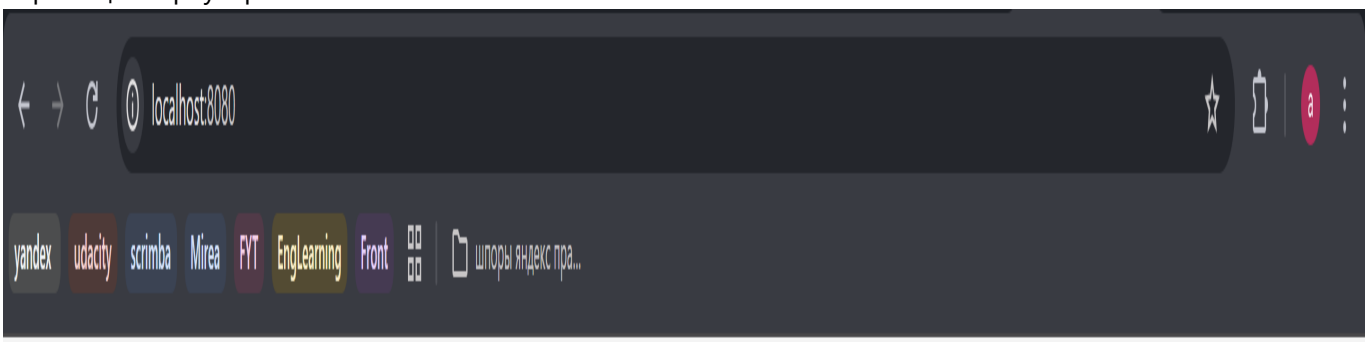
Запуск контейнера с nginx

```
alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1 (main)
$ docker run -d --name my-nginx-server -v my-nginx:/usr/share/nginx/html -p 8080:80 nginx:alpine
34a2f33bd14bb06474bd86f1f301709f9774ea2b7b4af381d94f9f21ab2e97fe
```

Копирование файла index.html внутрь контейнера

```
alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1 (main)
$ docker cp index.html my-nginx-server:/usr/share/nginx/html
Successfully copied 3.58kB to my-nginx-server:/usr/share/nginx/html
```

Страница в браузере



EAL 16

Информация о сервере:

Время на сервере: 10.03.2026, 22:32:46

IP-адрес сервера:

Имя сервера:

Тестовая страница Nginx • Создано для проверки конфигурации

/usr/share/nginx/html/index.html

Работа с сетью

Создание сети

```
alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1 (main)
$ docker network create my-app-network
267b564f32123ce77529b0517174ffbe41140212fe61ed95bf4d993124a7a682
```

Просмотр сетей

```
alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1 (main)
$ docker network ls
NETWORK ID     NAME            DRIVER    SCOPE
e3245d65c46e   bridge          bridge    local
382f94712bd9   docker_default  bridge    local
c5192df9021a   host            host      local
267b564f3212   my-app-network  bridge    local
f08968a2d103   none            null      local
```

Запуск контейнеров в одной сети

```
alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1 (main)
$ docker run -d --name postgres --network my-app-network -e POSTGRES_PASSWORD=secret -e POSTGRES_DB=myapp postgres:15-alpine
7f7c348a424fc63a33d3b528d9d775c0336c5b401419d39610219a8accd3b8a7
```

```
alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1 (main)
$ docker run -d --name app --network my-app-network -p 8080:80 nginx:alpine
e07ea40c0ad69f5315bebabb0623fddd029fdfde535fb6736c66af853a24cff0
```

Проверка связи между контейнерами

```
alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1 (main)
$ docker exec app ping postgres
PING postgres (172.19.0.2): 56 data bytes
64 bytes from 172.19.0.2: seq=0 ttl=64 time=0.518 ms
64 bytes from 172.19.0.2: seq=1 ttl=64 time=0.167 ms
64 bytes from 172.19.0.2: seq=2 ttl=64 time=0.098 ms
64 bytes from 172.19.0.2: seq=3 ttl=64 time=0.094 ms
```

Практическое задание

Создание сети my-bridge-network

```
alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1 (main)
$ docker network create --driver bridge my-bridge-network
089a416d6d7a96eef371818f6cd1757f2db952bcf74192fff063617172730e45
```

Запуск контейнеров postgres и nginx в одной сети и проверка их взаимодействия:

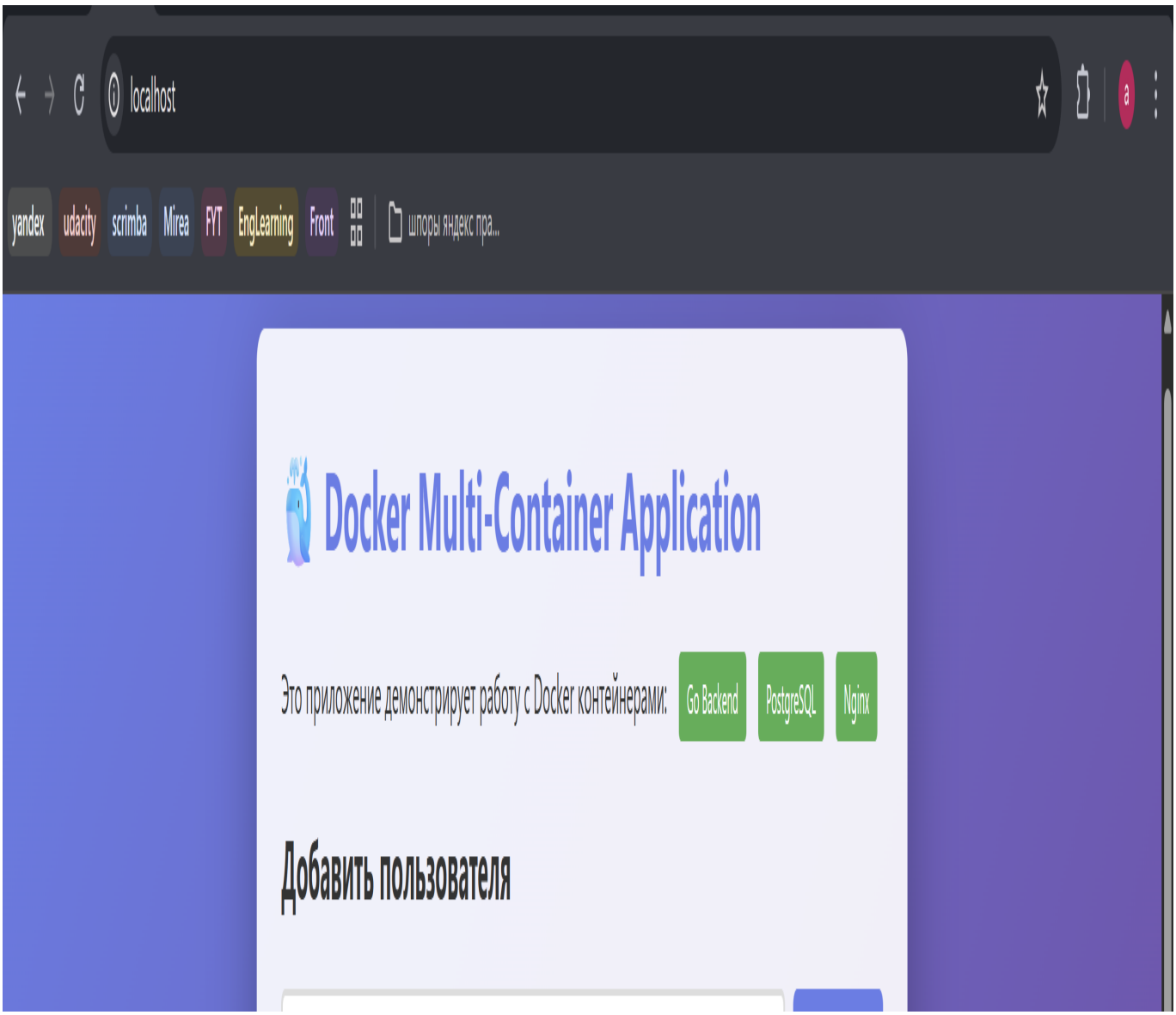
```
alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1 (main)
$ docker run -d --name my-postgres --network my-bridge-network -e POSTGRES_PASSWORD=secreet -e POSTGRES_DB=my-app postgres:15-alpine
9fb827d83a07cdba38db5190273e37f9bc98c0dbd74188206df77ec412b862d1
```

```
alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1 (main)
$ docker run -d --name my-nginx --network my-bridge-network -p 8080:80 nginx:alpine
5dd6e28e9cfd37bed04ebe4c67f60983b07e21a2655adb76918d21f860ba414c
```

```
alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1 (main)
$ docker exec my-postgres ping my-nginx
PING my-nginx (172.20.0.3): 56 data bytes
64 bytes from 172.20.0.3: seq=0 ttl=64 time=0.338 ms
64 bytes from 172.20.0.3: seq=1 ttl=64 time=0.116 ms
64 bytes from 172.20.0.3: seq=2 ttl=64 time=0.142 ms
64 bytes from 172.20.0.3: seq=3 ttl=64 time=0.123 ms
64 bytes from 172.20.0.3: seq=4 ttl=64 time=0.103 ms
```

2. Скриншоты работающего приложения

2.1 Главная страница



Введите имя

Добавить

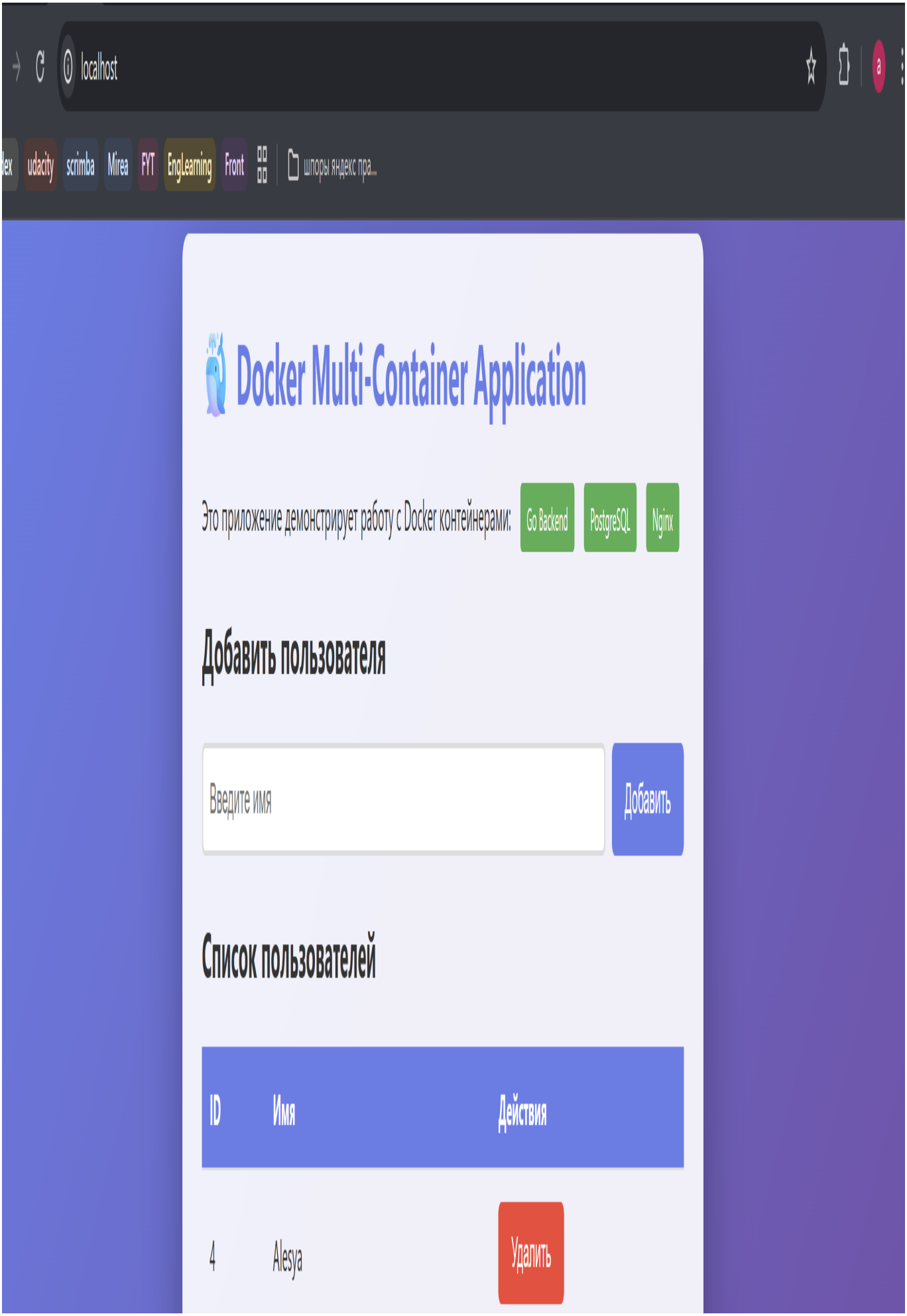
Список пользователей

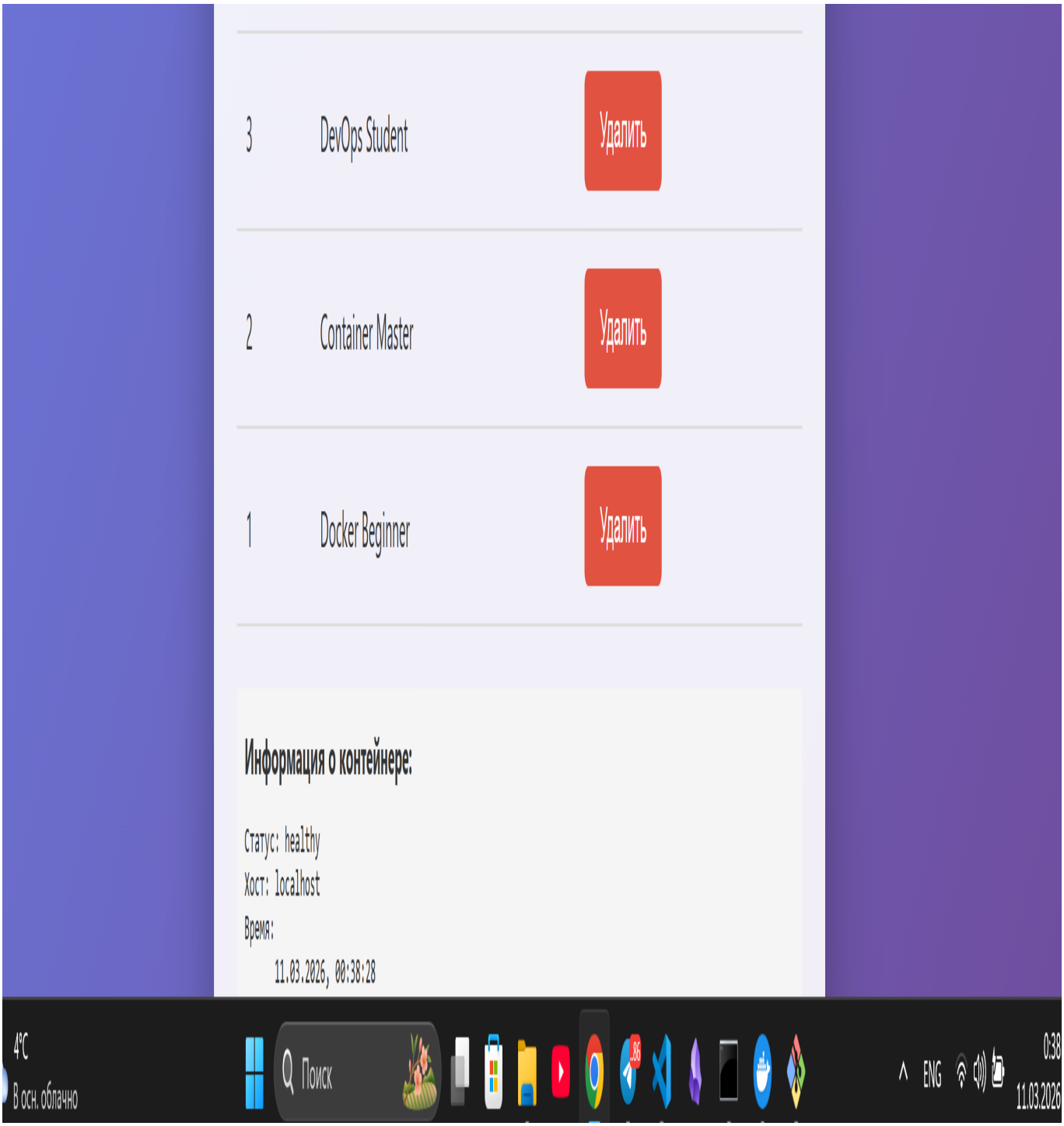
ID	Имя	Действия
3	DevOps Student	Удалить
2	Container Master	Удалить
1	Docker Beginner	Удалить

Информация о контейнере:

Статус: healthy
Хост: localhost
Время:
11.03.2026, 00:37:23

2.2 Добавление пользователя





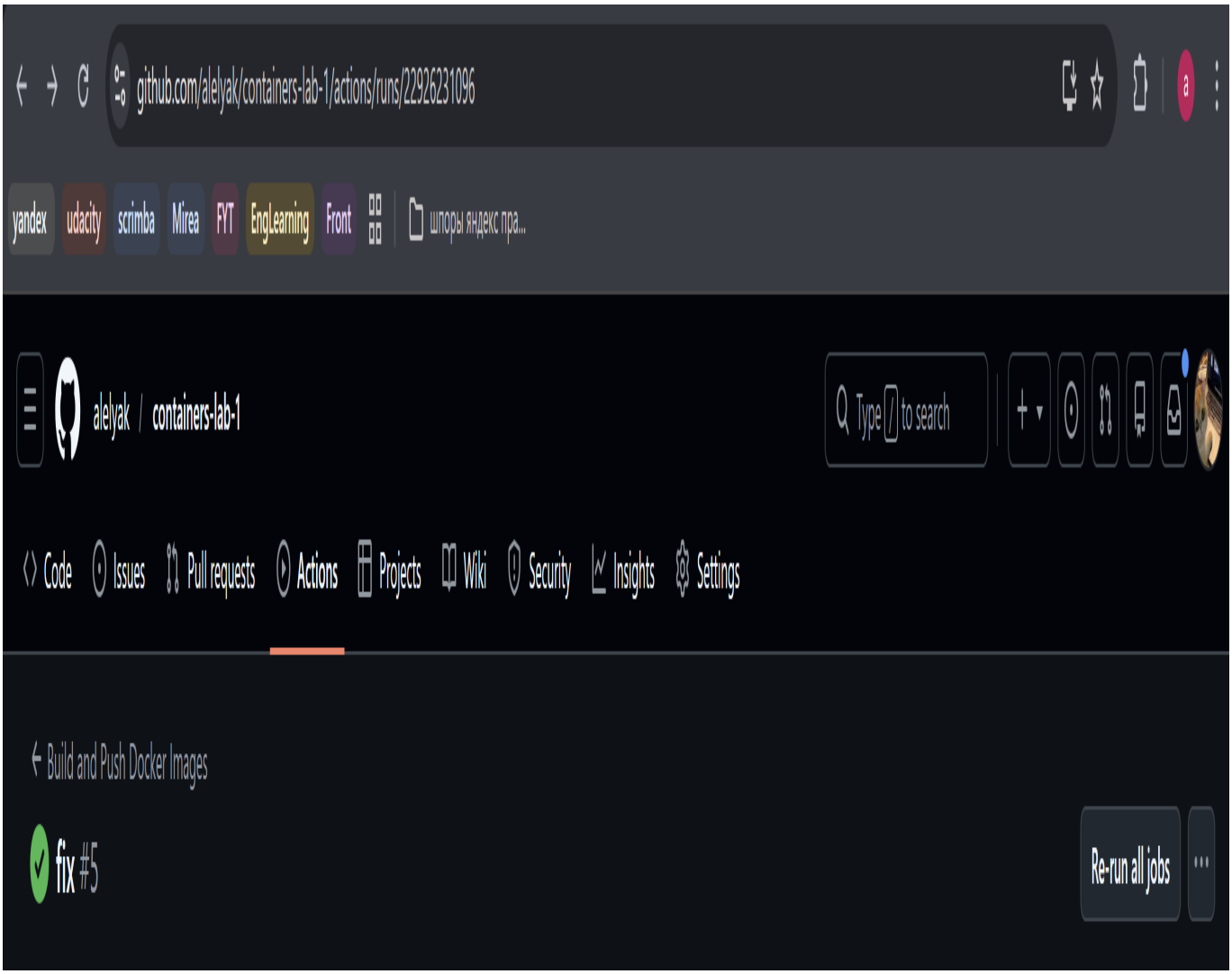
2.3 Список пользователей в БД

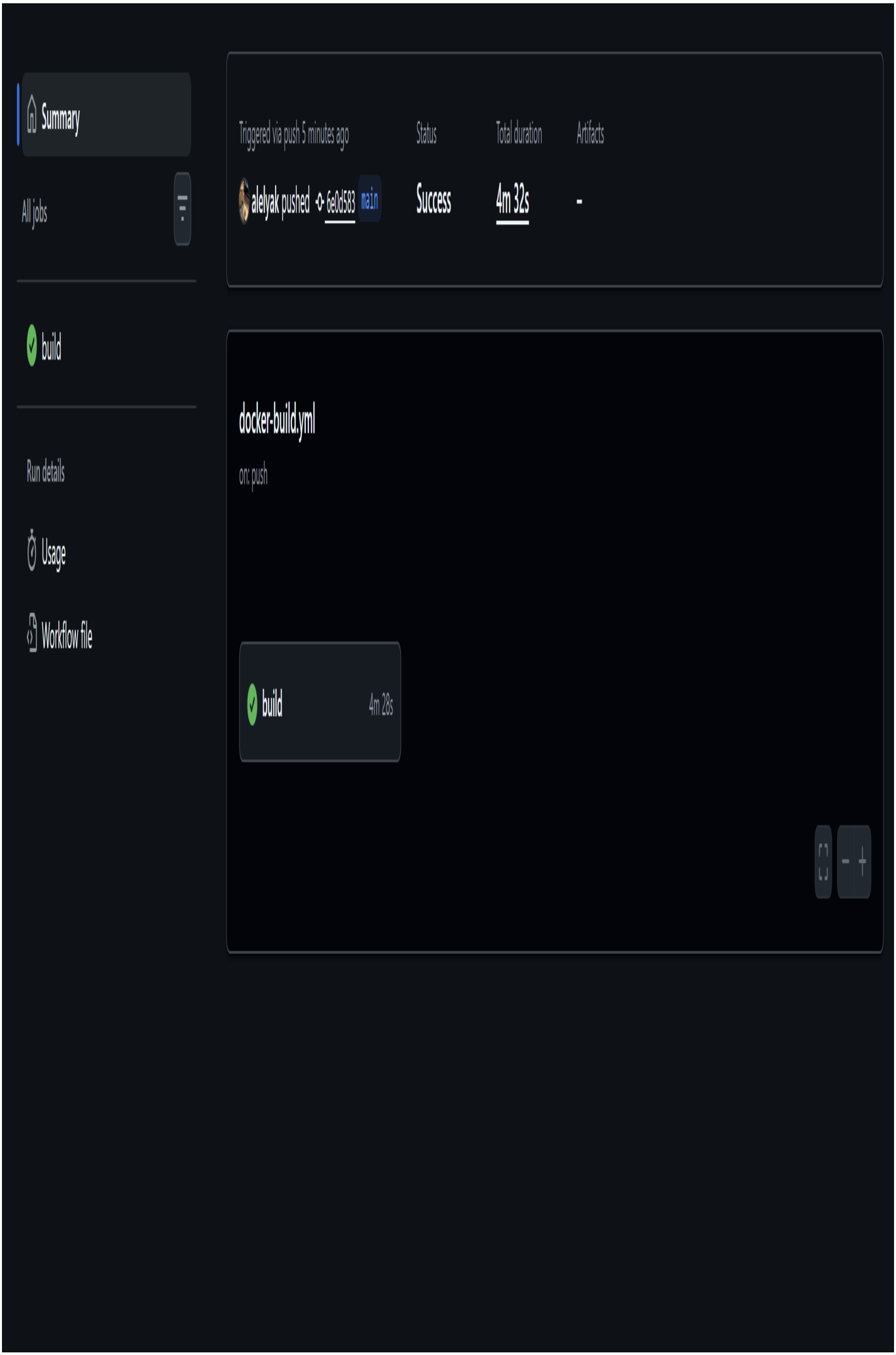
```
alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1/app (main)
$ docker exec -it lab1-postgres psql -U postgres -d myapp -c "SELECT * FROM users"
 id |       name       |         created_at
----+------------------+----------------------------
  1 | Docker Beginner  | 2026-03-10 21:27:09.650382
  2 | Container Master | 2026-03-10 21:27:09.650382
  3 | DevOps Student   | 2026-03-10 21:27:09.650382
  4 | Alesya           | 2026-03-10 21:38:29.162841
(4 rows)


What's next:
    Try Docker Debug for seamless, persistent debugging tools in any container or image → docker debug lab1-postgres
    Learn more at https://docs.docker.com/go/debug-cli/
```

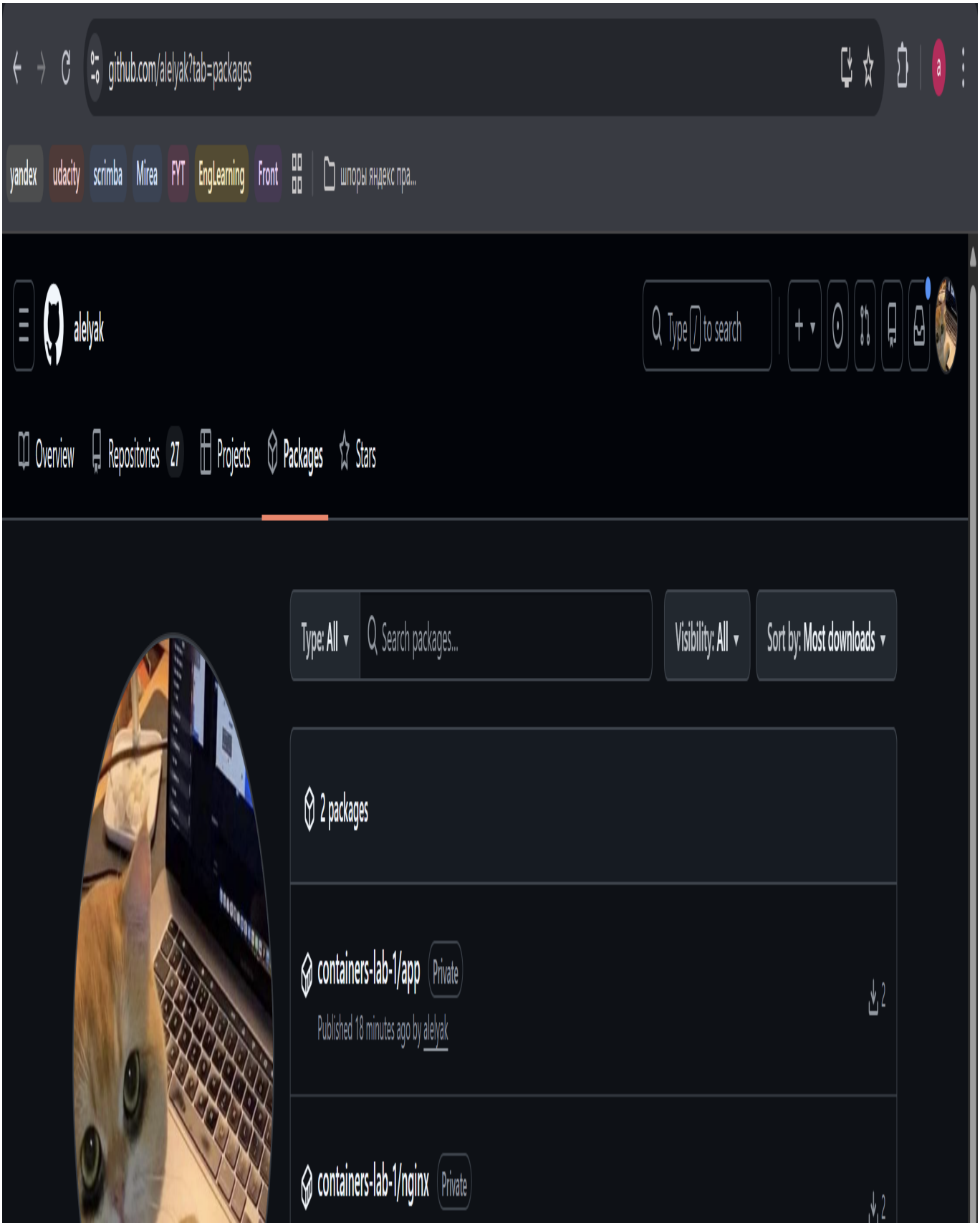
3. GitHub Actions

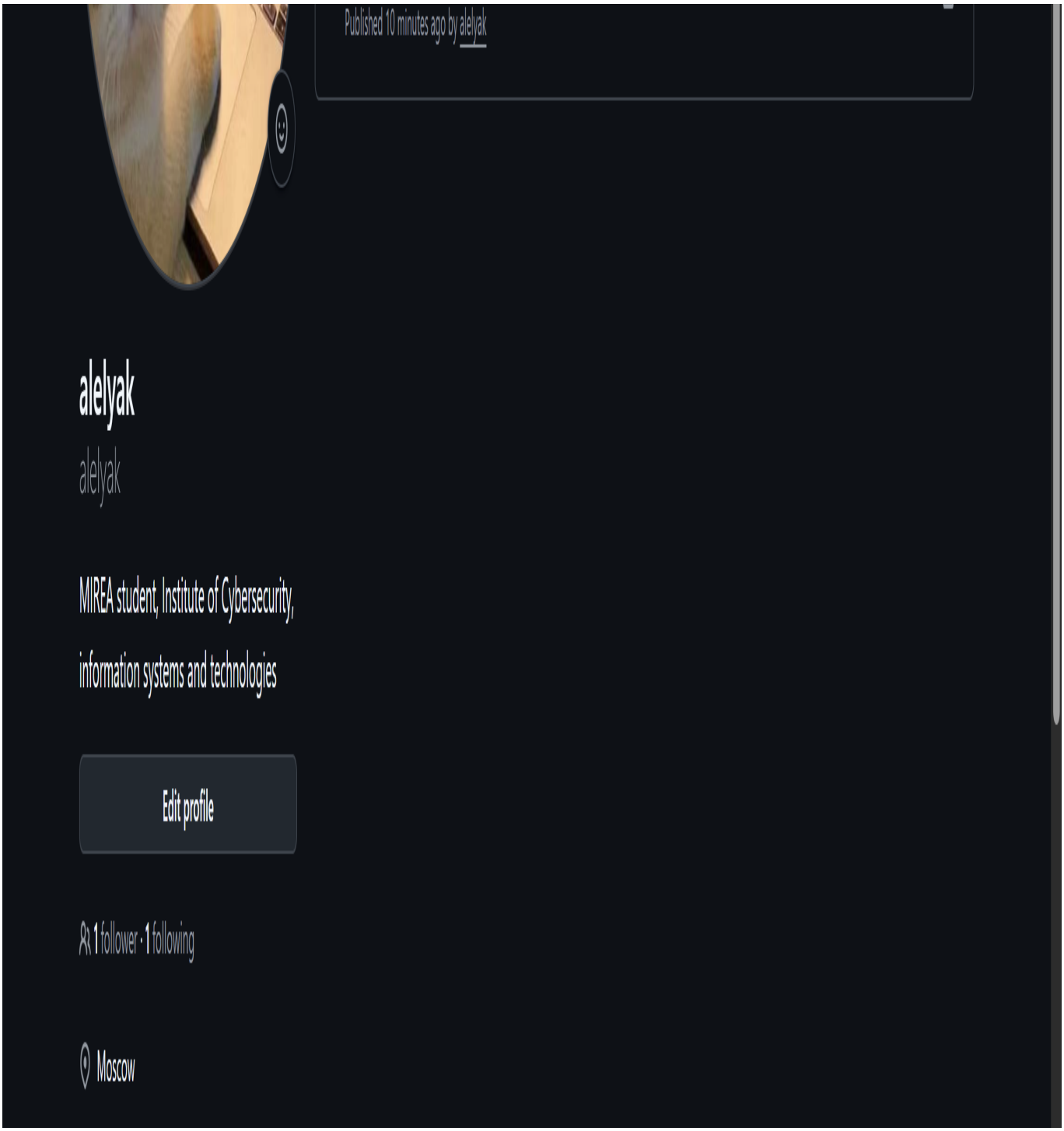
3.1 Успешный запуск workflow




Summary
All jobs
build
Run details
Usage
Workflow file
Triggered via push 5 minutes ago
Status
Total duration
Artifacts
alelyak pushed 6e0d583 main
Success
4m 32s
-
docker-build.yml
on: push
build
4m 28s

3.2 Опубликованные образы в GHCR





4. Выводы

В ходе выполнения практической работы я: освоила основные команды Docker для управления контейнерами, образами и томами, научилась писать Dockerfile с multi-stage сборкой, создала многокомпонентное приложение на Go с PostgreSQL и Nginx, настроила CI/CD пайплайн в GitHub Actions для автоматической сборки образов, успешно опубликовала образы в GitHub Container Registry (GHCR). Основные трудности, с которыми столкнулась: конфликты портов с уже запущенными сервисами, настройка GitHub Actions и исправление ошибок в workflow